📐 Grade 12 Mathematics – Algebra

CAPS aligned · Study notes for final exams · Exponents, quadratics, sequences & more

1. Exponents and Surds

📌 Exponent laws (revision)

$a^m \times a^n = a^{m+n}$	$a^m \div a^n = a^{m-n}$
$(a^m)^n = a^{mn}$	$(ab)^n = a^n b^n$
$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$	$a^{-n} = \frac{1}{a^n}$
$a^0 = 1 \;(a \neq 0)$	$a^{\frac{m}{n}} = \sqrt[n]{a^m}$

Example: Simplify $(2x^3y^{-2})^2$
$= 2^2 x^{6} y^{-4} = \frac{4x^6}{y^4}$

🔹 Surds (radicals)

- $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$
- $\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$
- $\sqrt{a^2} = |a|$
- **Rationalising denominator:** $\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$
- For binomial: $\frac{1}{\sqrt{a}+\sqrt{b}} = \frac{\sqrt{a}-\sqrt{b}}{a-b}$

2. Quadratic Equations

General form: $ax^2 + bx + c = 0,\; a \neq 0$

🔹 Methods of solving

- **Factorisation** – set each factor = 0
- **Completing the square** – create perfect square trinomial
- **Quadratic formula** – $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

Example (formula): $2x^2 - 3x - 2 = 0$
$a=2,\; b=-3,\; c=-2$
$x = \frac{3 \pm \sqrt{9+16}}{4} = \frac{3 \pm 5}{4}$
$x = 2 \quad \text{or} \quad x = -\frac{1}{2}$

🔹 The discriminant $\Delta = b^2-4ac$

Discriminant	Nature of roots
$\Delta > 0$	Two distinct real roots
$\Delta = 0$	One real root (double root)
$\Delta < 0$	No real roots (complex)

3. Quadratic Inequalities

- Rewrite as $ax^2+bx+c \; \gt/\lt \; 0$
- Solve the corresponding equation $ax^2+bx+c=0$
- Sketch parabola (opens up if $a>0$)
- Read solution set from the graph

Example: $x^2 - 5x + 6 < 0$
Roots: $x=2,\; x=3$
Parabola opens up → below x-axis between roots → $x \in (2,3)$

4. Simultaneous Equations (Linear + Quadratic)

Method: substitution – solve linear for one variable, plug into quadratic.

$\begin{cases} y = x+2 \\ x^2 + y^2 = 10 \end{cases}$
Substitute: $x^2 + (x+2)^2 = 10$ → $2x^2+4x-6=0$ → $x^2+2x-3=0$
$(x+3)(x-1)=0$ → $x=-3$ or $x=1$ → $y=-1$ or $y=3$
Solutions: $(-3,-1)$ and $(1,3)$.

5. Arithmetic & Geometric Sequences

📈 Arithmetic sequence

$T_n = a + (n-1)d$ $S_n = \frac{n}{2}[2a+(n-1)d]$

Example: $3,7,11,15,\dots$ → $a=3,\; d=4$ → $T_n = 4n-1$
$S_{10} = \frac{10}{2}[2\times3 + 9\times4] = 5\times(6+36)=210$

📉 Geometric sequence

$T_n = a r^{n-1}$ $S_n = \frac{a(r^n-1)}{r-1} \; (r \neq 1)$
Sum to infinity (if $|r|<1$): $S_\infty = \frac{a}{1-r}$

Example: $2,6,18,54,\dots$ → $a=2,\; r=3$ → $T_5 = 2\times3^{4}=162$
$S_6 = \frac{2(3^6-1)}{3-1} = \frac{2(729-1)}{2}=728$

🔹 Sigma notation $\sum$

$\sum_{k=m}^{n} f(k)$ means sum of $f(k)$ from $k=m$ to n.

$\sum_{k=1}^{4}(2k+1) = 3+5+7+9 = 24$

6. Practice Problems

1. Solve $2x^2 -5x -3 = 0$ by factorisation.
2. Given $x^2 + 2kx + 9 = 0$ has equal roots, find k.
3. Solve inequality: $-x^2+4x+5 > 0$.
4. Solve simultaneously: $y = 2x-1$ and $x^2+y^2=10$.
5. For the sequence $5,9,13,\dots$ find T_{20} and S_{20}.
6. Geometric sequence: $T_3=24$ and $T_6=192$. Find a and r.

📌 Quick solutions

1. $(2x+1)(x-3)=0$ → $x=-\frac12$ or $x=3$
2. $\Delta = 4k^2-36=0$ → $k=\pm3$
3. Multiply by -1: $x^2-4x-5<0$ → roots $-1,5$ → solution \(-1
4. Substitute → $5x^2-4x-9=0$ → $x=1.8$ or $x=-1$; points $(1.8,2.6),\;(-1,-3)$
5. $a=5,d=4$ → $T_{20}=5+19\times4=81$; $S_{20}=10\times(5+81)=860$
6. $ar^2=24,\; ar^5=192$ → divide: $r^3=8$ → $r=2$, then $a=6$

7. Exam Tips

- ✔️ Check restrictions: denominator ≠ 0, expression under square root ≥ 0.
- ✔️ Use discriminant for “real”, “rational” or “equal” roots questions.
- ✔️ For sequences, always write down a, d (or r) first.
- ✔️ Sigma notation: write first and last terms to avoid errors.
- ✔️ In inequalities, never multiply by an expression that could be negative — keep sign analysis.

💡 **Memory aid**
Quadratic formula song: “x equals minus b, plus or minus square root, of b squared minus 4ac, all over 2a.”

📖 Practice with past NSC papers & memos. Master algebra → build foundation for calculus & functions.
Good luck with your matric exams! 🚀